

Ahash Thirukumaran

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Education

New Jersey Institute of Technology - *B.S. in Computer Science*

May 2026

GPA: 3.3

- **Relevant Coursework:** ML, Computer Architecture, Data Science, Computational Theory, Data Structures & Algorithm, Software Engineering Design, Intensive Programming in Linux

Experience

Software Engineering Intern | *Wakefern Food Corp. Tech Division*

Feb 2026 – May 2026

- Building a full-stack **iOS/Android** event app for the ShopRite LPGA Classic with **8 major features** (auth, scheduling, leaderboard, gallery, notifications, sponsor management, interactive map, admin panel) across a dual **Ionic mobile + Angular web** frontend (**41 pages/components**); launching May 15, 2026.
- Architected the backend using **Spring Boot** and containerized the development environment with **Docker** to ensure consistent builds across the team.
- Designed a **13-entity PostgreSQL** schema supporting event scheduling, player 4-some assignments, sponsor tiers, and user profiles.
- Built **57 RESTful API endpoints** serving personalized schedules, tee-time data, real-time scores, and event content to an **Angular + Ionic** frontend.

Computer Science Course Assistant | *NJIT*

Jan 2024 – May 2026

- Tutored 100+ students in an Object-Oriented Programming course (**Java**), clarifying complex topics such as **inheritance, polymorphism, and advanced data structures**.
- Held regular office hours and group review sessions, significantly boosting average assignment completion rates.
- Evaluated and graded programming labs, projects, and exams, ensuring consistent application of grading standards.

Projects

Resume Tailor

- Built a full-stack application that parses PDF/DOCX resumes and tailors it to a given job description using **NLP**.
- Designed a **Python** microservice for resume parsing and integrated it with a **Node.js/Express** backend.
- Implemented inter-process communication via child processes and **RESTful API endpoints**.
- Developed a **React** frontend enabling secure resume uploads, validation, and tailored outputs.
- Automated testing, linting, and deployment using GitHub Actions **CI/CD** pipelines.

Covey.Town UI Feature Development

- Developed a coordinate-based virtual pet guide frontend using animated sprite sheets, real-time movement updates, and world-state synchronization to assist user navigation in Covey.Town's multiplayer environment.
- Collaborated on a 4-person **agile Scrum** team across 4 sprints, participating in sprint planning and stand-ups.
- Deployed via **AWS Lightsail** and maintained **CI/CD** pipelines to automate testing and deployments, reducing deployment errors by **60%** and enabling faster feature delivery.

Secure Sandbox Environment

- Built a lightweight container runtime from scratch in C++ using Linux kernel primitives (`clone()`, namespaces, `chroot`, `cgroups v2`, `seccomp`), achieving process isolation equivalent to Docker without any container framework.
- Designed and implemented a default-deny `seccomp` syscall filter, reducing the kernel attack surface to ~100 explicitly whitelisted calls and blocking all filesystem/namespaces escape vectors.
- Enforced layered resource limits via `cgroups v2` (50% CPU quota, 256 MB memory, 64 PID cap) with `setrlimit` as a guaranteed fallback, preventing sandboxed processes from impacting host stability.
- Managed the full privilege lifecycle, dropping root to a dedicated unprivileged user post-setup while selectively preserving `CAP_NET_RAW` via Linux ambient capabilities to maintain network functionality across `execve`.

Technical Skills

Languages: Java, Python, C, C++, SQL, JavaScript, HTML, TypeScript

Frameworks: React.js, Node.js, Express.js, Next.js, Spring Boot, Angular, Ionic

Databases: MongoDB, MySQL, PostgreSQL, Redis

Cloud & Tools: AWS, Git, GitHub, Linux/Unix, Docker